

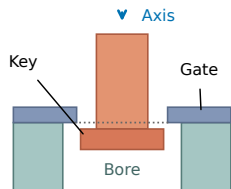
a Task geometry

Keyed peg and gate



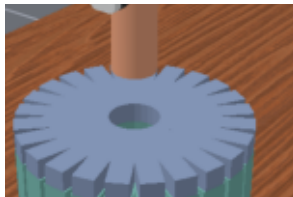
Actual simulation geometry

Key cleared: section



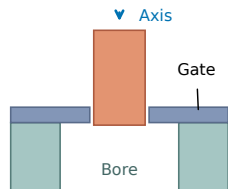
Schematic; yaw permitted below gate

Circular control



Actual simulation geometry

Circular gate: section



Schematic; no keyed yaw constraint

b Keyed-to-circular insertion

Align



Match the keyed opening

Enter



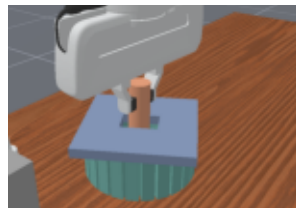
Key has not fully cleared

Clear / Unlock



Key below gate; yaw permitted

Insert



Continue into circular bore

c Representative task relations

Sliding



Drawer · state-level probe

Planar transport



Heading-constrained · closed-loop

Revolute



Microwave door · state-level probe

Support / placement



Bowl-stove · state-level probe